

| Question |     |      |     | Marking details                                                                                                                                                                                                                                                                                                                                                                                                          | Marks available |     |     |       |       |      |
|----------|-----|------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |     |                                                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5        | (a) | (i)  | I   | {is complimentary to/attaches to} the <u>start</u> of the sequence/<br>allows binding of {taq/DNA} polymerase                                                                                                                                                                                                                                                                                                            | 1               |     |     | 1     |       |      |
|          |     |      | II  | breaks hydrogen bonds between strands/<br>separating the two strands/<br>DNA becomes single stranded                                                                                                                                                                                                                                                                                                                     | 1               |     |     | 1     |       |      |
|          |     |      | III | to allow the primers to { bind / anneal} (to the DNA strands)                                                                                                                                                                                                                                                                                                                                                            | 1               |     |     | 1     |       |      |
|          |     | (ii) |     | thermally stable/can withstand {high temperatures/70°C+}/<br>resistant to heat/eq                                                                                                                                                                                                                                                                                                                                        | 1               |     |     | 1     |       |      |
|          | (b) |      |     | <b>Any two (×1) from:</b> <ul style="list-style-type: none"> <li>Populations are geographically isolated/ or description of (1)</li> <li>Different environments select different alleles/<br/>genetic drift /<br/>no gene flow (1)</li> <li>(In each area) they can interbreed and so {remain genetically close/ gene pools remain similar}/ ORA (1)</li> </ul>                                                          |                 | 2   |     | 2     |       |      |
|          | (c) |      |     | Elephants with tusks would be (more likely to be) {poached/killed}/<br>greater <u>selection pressure</u> on tusked elephants/ ORA (1)<br><br>{Poaching/killing} is selecting {for tuskless elephants/against<br>tusked}/<br>tuskless elephants have a <u>selective</u> advantage/ ORA (1)<br><br>so {pass on their {alleles/ mutation} (for no tusks)/<br>increased frequency of tuskless alleles in gene pool / ORA (1) |                 | 1   | 2   | 3     |       |      |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                            | Marks available |          |          |           |          |          |
|----------|-----|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                            | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          | (d) |  |  | Increased population size (1)<br>because tusk-less animals are not {poached/killed} (for ivory) (1)<br><br><b><u>OR</u></b><br><b><u>Award two marks for:</u></b> <ul style="list-style-type: none"><li>Decreased population size because {elephants cannot defend against predators/feed as well} without tusks</li></ul> |                 | 2        |          | 2         |          |          |
|          | (e) |  |  | <b>Any two (×1) from:</b> <ul style="list-style-type: none"><li>Identify {where it has come from /population source / group} (1)</li><li>Ref to Anti-poaching action targeting populations at risk (1)</li><li>Identify recent illegal ivory (1)</li></ul>                                                                 |                 | 2        |          | 2         |          |          |
|          |     |  |  | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                    | <b>4</b>        | <b>7</b> | <b>2</b> | <b>13</b> | <b>0</b> | <b>0</b> |